ALA-2

Topic– User defined exception

Q-1. What is a user-defined exception in Java?

Ans. A user-defined exception is a custom exception created by the programmer by extending the Exception class. It is used to handle specific errors that are not covered by built-in exceptions.

Q-2. Why do we use user-defined exceptions?

Ans. We use user-defined exceptions to handle application-specific errors and improve code readability and debugging.

Q-3. How do you create a user-defined exception?

Ans. By creating a class that extends Exception.

Example:

```
Class MyException extends Exception {
  public MyException(String message) {
    super(message);
  }
}
```

Q-4. How do you throw a user-defined exception?

Ans Using the throw keyword.

Example:

```
throw new MyException("Error occurred");
```

Q-5 How do you handle a user-defined exception?

Ans. Using try-catch block.

Example:

```
try {
   throw new MyException("Error");
} catch (MyException e) {
   System.out.println(e.getMessage());
}
```

Q-6. What is the difference between throw and throws?

Ans. throw → used to actually throw an exception

throws → used in method declaration to declare exceptions

Q-7. Can user-defined exceptions be checked or unchecked?

Ans. Checked → extend Exception

Unchecked → extend RuntimeException

Q-8. Give an example of user-defined exception for divide by zero.

Ans.
```
class DivideByZeroException extends Exception {
public DivideByZeroException(String msg) {
super(msg);
}
}

class Test {
public static void main(String[] args) {
try {
int b = 0;
if (b == 0)
throw new DivideByZeroException("Cannot divide by zero");
} catch (DivideByZeroException e) {
System.out.println(e.getMessage());
}
}
}
```

Q-9. What are advantages of user-defined exceptions?

Ans. Better error handling

Custom messages

Improves debugging

Makes code more meaningful

Q-10. Can we use multiple constructors in user-defined exception?

Ans. Yes, we can define multiple constructors just like normal classes.

Example:

```
class MyException extends Exception {
  public MyException() {}
  public MyException(String msg) {
    super(msg);
  }
}
```